

Computer Networking

The 80/20 Revision Guide — 2025/26 syllabus

Part IB — Cambridge CST — Paper 5

How this guide is weighted (read first)

New lecturer / exam setter this year. This guide is built on the **lecturer’s own 2025/26 fine-grained syllabus and clarifications**, *not* on past-paper frequency — with a new setter, old papers are an unreliable guide to emphasis and question style. The governing principle the lecturer gave is: **everything lectured is examinable; anything marked “not lectured 25/26” is out.** Textbook reading is the relevant sections of *Kurose & Ross* and *Peterson & Davie* cited per topic in the course pages.

NOT examinable in 25/26 — do not revise these

These appear in old papers / the supervision sheet but were **not lectured this year** — skip them:

- **TCP 3-way handshake / connection establishment & teardown**; the “TCP equation” window integration; buffer-sizing theory.
- The **four TCP ARQ timers** and **silly window syndrome** (worksheet Q22).
- **DNSSEC** and **Secure DNS**; cookies; dark web; P2P file sharing; email; **DASH**.
- **IPv6 details** — Neighbour Discovery, **SLAAC** (only the first two IPv6 slides count); **IP multicast**; **IP fragmentation**.
- Physical-layer **block diagrams / power budgets**; GF(2) arithmetic detail of CRC; HomePlug; Bluetooth; data-centre networks; Viterbi; AutoNET.
- Specific **iterative-vs-recursive DNS resolution** patterns (see the depth note under Application).

New emphasis this year

Emphasised this year but light/absent in old papers — make sure you can discuss these: **QUIC / HTTP 2,3** (the *roles*: avoiding head-of-line blocking, avoiding round-trips, security); **CUBIC** and **ECN**; **CDN** (three approaches, Akamai-style overlay, eager push, reverse vs forward proxies); **fair queuing / fairness**; **VLAN, VPN, tunnelling** and “layer-order violations”; **spanning tree**.

1 Transport layer (Tier 1)

Exam-critical

Core functions: **port multiplexing/demultiplexing, reliable delivery, end-to-end flow control, congestion avoidance**. UDP = thin (ports + checksum, connectionless); TCP = reliable bytestream over IP’s narrow waist.

Reliable conveyance: sequence numbers + acks + timeouts; **Go-Back-N** (cumulative ack, retransmit window) vs **Selective Repeat** (per-packet ack/buffer) — know the trade-off (SR less wasteful, more state).

Sliding window: sender limited by $\min(\text{cwnd}, \text{rwnd})$ — congestion window vs receiver window.

Exam-critical

[title=Congestion control] **AIMD** — additive increase (+1 MSS/RTT), multiplicative decrease ($\times \frac{1}{2}$ on loss); this sawtooth is what makes competing flows converge to a **fair** share. **Slow start** — cwnd grows *exponentially* (doubles per RTT) up to *ssthresh*, then AIMD. **Fast recovery** on triple-duplicate-ack. **CUBIC** — window a cubic function of time since the last loss (probes high-bandwidth links faster than classic AIMD). **ECN** — routers *mark* packets (instead of dropping) to signal congestion early.

Worth knowing

RTT & jitter measured by the sender (used to set timeouts). **Fairness:** one definition lectured (e.g. max-min / equal share); **fair queuing** enforces it in the router rather than relying on cooperative end-hosts. End-to-end vs hop-by-hop reliability (the end-to-end argument). **TLS** — key points only, no protocol detail.

Exam trap

The **3-way handshake is OUT this year** — don't lead a TCP answer with SYN/SYN-ACK/ACK. Focus on *reliability mechanisms* (SR/GBN, sliding window) and *congestion control* (AIMD/slow-start/CUBIC/ECN), which is where the marks now are.

2 Link layer & LANs (Tier 1)

Exam-critical

Framing and **MAC addressing** (flat 48-bit, local scope). **Error detection:** *checksums* and *CRC* (know what they do; **no GF(2) arithmetic detail needed**); brief **Hamming-distance** error *correction*; erasure channels.

Media access control on a shared medium — three families: *turn-taking*, *polling*, *contention*.

Exam-critical

[title=Multiple-access protocols] **ALOHA** (transmit anytime; ~18% max efficiency) → **slotted ALOHA** (~37%) → **CSMA** (listen before talk) → **CSMA/CD** (Ethernet: detect collisions, jam, **exponential backoff** with a growing *contention window*).

Wireless can't detect collisions well: **hidden** and **exposed terminal** problems ⇒ **RTS/CTS** handshake (collision *avoidance*, the basis of Wi-Fi/CSMA/CA).

Worth knowing

Switched L2: migration from shared medium to learning **switches**; the **spanning-tree** protocol prevents loops in a bridged topology. **VLAN** — several virtual LANs over one physical LAN (security/convenience, cheaper than an air gap; needs configuration). The PHY→DLL interface (Tx/Rx clock, data, frame-start, RSSI/carrier-detect) and whether a link is **duplex vs half-duplex** (duplex = a pair of simplex channels) matter; physical-layer block diagrams do not.

3 Network layer (Tier 1)

Exam-critical

[title=Addressing & forwarding] **Naming vs addressing vs routing vs forwarding. IP addressing, subnetting, and longest-prefix match** at the router. Router block diagram and the **trade-offs of where to put buffers. ICMP** and **traceroute**; **ARP** (IP→MAC on a LAN). IPv6: *first two slides only* — exists, larger addresses; **no Neighbour Discovery/SLAAC detail**.

Exam-critical

[title=Routing] **Forwarding** (per-packet, local table lookup) vs **routing** (computing the tables). Intra-domain protocols:

- **Link State** (e.g. OSPF): each router floods link costs, then runs *Dijkstra* on the full map.
- **Distance Vector** (e.g. RIP): *Bellman–Ford*, exchange distance estimates with neighbours; watch count-to-infinity.

Know the response to disruptions, and **hierarchic vs flat** routing (why the Internet is hierarchical).

Worth knowing

[title=Middleboxes — the “four-sentence” depth the lecturer asked for]

- **NAT** — a middlebox with the *same* network layer each side; translates local IPs (10.x / 192.168.x) to one/few public IPs using the **port space**.
- **Firewall** — refuses connections per security rules (e.g. no incoming TCP); commonly *bundled* with NAT.
- **VPN** — **tunnelling** / deliberate layer-order violation to give the illusion of being on a local LAN while elsewhere.
- **Gateway** — a per-application middlebox *above* the network layer, with *different* network layers each side (vs the old “border gateway router”, which is just a router between autonomous domains).

DHCP (examinable, core idea only): dynamic allocation of IP addresses from a *pool* (vs manual), over a **broadcast subnet**; a MAC usually gets the *same* address as before within a 1–2 week timeout. *Not* SLAAC.

4 Application layer (Tier 1 — big new emphasis)

Exam-critical

[title=DNS] A distributed, **hierarchic** system of **caches and delegated authorities** (with many ad-hoc local edits). Know the **resource-record 4-field structure** (name, type, value, TTL) and the record **types: A, MX, CNAME, NS**. `gethostbyname`. **Load balancing** via **anycast** (and how anycast can be abused by filling routing tables to steer queries to a local/ISP-preferred server), caching, and **soft state**.

Exam trap

For DNS the lecturer does **not** want memorised iterative-vs-recursive resolution patterns. He wants: (1) the purpose of the record types; (2) the core principle (hierarchy of caches + delegated authorities); (3) the ability to *reason about* feasible resolution/anycast use-patterns yourself. **DNSSEC is out.**

Exam-critical

[title=HTTP and content delivery] **HTTP 1.0 vs 1.1** (persistent connections); **conditional GET** (If-Modified-Since $\rightarrow$ 304). **Web caches/proxies** — *forward* (near client) vs *reverse* (near server) positioning; eager **push** of content. **CDN** — the *three distinctive approaches*, the Akamai-style **overlay network**, and multiple sites per server.

QUIC / HTTP 2,3 (numbering unimportant; *roles* are): stream multiplexing to avoid **head-of-line blocking**, **0-RTT** setup to avoid round-trips, and built-in **security/encryption** (runs over UDP).

Worth knowing

Multimedia: P2P vs client/server; CBR vs VBR (JPEG/MPEG familiarity assumed); **dynamic playout-buffer sizing** to absorb jitter; SIP mentioned. (DASH is *out*.)

5 Foundations & architecture (Tier 2)

Exam-critical

[title=Foundations] **Sharing/multiplexing:** spatial (SDM), **TDM**, **FDM**, **WDM**. **Directionality:** simplex / duplex / half-duplex; **cast modes:** unicast / multicast / broadcast / anycast. **Switching:** circuit vs packet; connection-oriented vs connectionless. **Traffic:** mean vs peak rate, burstiness, and **statistical multiplexing gain** (why packet switching wins for bursty traffic). **Performance:** speed of light, **RTT**, **bandwidth–delay product**, utilisation, queuing delay.

Worth knowing

[title=Architecture & philosophy] **Modularity, layering, abstraction;** the **OSI** reference model; and crucially *when layering is harmful* (it can hide information needed for performance). The **Internet narrow waist** (everything over IP). The **end-to-end argument** vs **middle-boxes**. First mention of **VLANs** and **tunnelling** as deliberate layering violations.

Worth knowing

[title=Physical layer (Tier 3 — minimal for CS)] The lecturer notes most L1 detail is *engineering*, not CS. Know only: the basic **channel-capacity** idea (Shannon $C = B \log_2(1 + S/N)$; Nyquist for a noiseless channel), **cable vs wireless** differences, the **names/distinguishing features** of the schemes presented, the **PHY→DLL interface signals** (Tx/Rx clock, data, frame-start, write-gate/power, RSSI, coding-violation/FEC metrics), **duplex vs half-duplex**, and multi-channel support (CDMA/FDM). *Skip* block diagrams and power budgets.

Exam checklist

Exam-critical

1. **Transport:** mux/demux, UDP vs TCP; GBN vs SR; sliding window = $\min(\text{cwnd}, \text{rwnd})$; **AIMD** (sawtooth $\rightarrow$ fairness), slow-start, fast recovery, **CUBIC**, **ECN**; fair queuing. *Not* the 3-way handshake.
2. **Link:** framing, MAC addresses, CRC/checksum (no GF(2)); ALOHA $\rightarrow$ slotted $\rightarrow$ CSMA $\rightarrow$ CSMA/CD (backoff); hidden/exposed terminals $\rightarrow$ RTS/CTS; switches + spanning tree; VLAN.
3. **Network:** subnetting + longest-prefix match; router buffers/trade-offs; ICMP/traceroute/ARP; **DHCP** (pool, broadcast, same-addr-within-timeout); routing **Link State (OSPF/Dijkstra)** vs **DV (RIP/Bellman-Ford)**; hierarchic routing. IPv6 exists (no SLAAC/ND).
4. **Middleboxes (4-sentence depth):** NAT (port translation, same L3 each side), firewall (rule-based refusal, bundled with NAT), VPN (tunnelling/layer violation), gateway (per-app, different L3 each side).
5. **Application:** DNS (record types A/MX/CNAME/NS, hierarchy of caches + delegation, anycast, soft state — *not* resolution patterns, *not* DNSSEC); HTTP 1.0/1.1, conditional GET, forward/reverse proxies, **CDN** (3 approaches, overlay, push); **QUIC/HTTP2-3** (avoid HOL blocking, avoid round-trips, security).
6. **Foundations:** TDM/FDM/WDM, cast modes, circuit vs packet, **statistical multiplexing gain**, RTT & **bandwidth-delay product**; layering/OSI/narrow-waist/end-to-end, when layering hurts.